

Final Report
Remote Car Control via a Driving Simulator

Project Manager Documentation: Weekly Activity Reports

Prepared by: Lili Nguyen

Sponsor: Dr. Bin Xu
Instructor: Dr. Cliff Fitzmorris

University of Oklahoma
ECE 4773: Capstone Design
Spring 2026

Table of Contents

Week 1	2
Week 3	9
Week 4 and 5	11
Week 6	14
Week 7.....	16
Week 8.....	18
Week 9.....	20
Week 10.....	22

Week 1

2/16/26 to 2/22/26

Executive Summary

This week, the team completed the initial system-level teleoperation architecture for the project. The main control path was defined using the Logitech G29 steering wheel, a laptop for input capture, MQTT communication, and an ESP32 for vehicle-side control. The team also began preparing software components for the driving simulator, ESP32 integration, controller data mapping, and PDR documentation.

Next week, the team will focus on setting up MQTT communication, testing input capture from the Logitech G29, bench-testing ESP32 control, and finalizing the diagrams needed for the PDR.

Progress Toward Objectives and Assignments Completed:

Ousmane Segba Kone (Total Time Spent: 4 hours)

- (1.5 hours) Finalized the overall teleoperation system architecture.
- (1.5 hours) Identified the required software components for the system, including:
 - Python script for reading Logitech G29 inputs
 - MQTT broker for message routing
 - ESP32 code for receiving commands and controlling outputs
- (1 hour) Defined the control message format, including:
 - Steering
 - Throttle
 - Brake
 - E-stop

Pundit Vorakitolan (Total Time Spent: 3.5 hours)

- (1 hour) Reviewed project documentation.
- (2 hours) Reviewed the Pygame library documentation and wrote preliminary test code for the driving simulator.
- (0.5 hour) Prepared for the PDR.
- Confirmed the expected interface scope with the sponsor:
 - Driving simulator outputs acceleration and turning data
 - CAN bus commands can be disregarded
 - Main focus is sending serial commands through the ESP32

Van Lang (Total Time Spent: 3 hours)

- (1 hour) Set up the Arduino IDE, installed ESP32 and Bluepad32 libraries, and installed required drivers.
- (1 hour) Researched driver interfaces and vJoy virtual joystick support.
- (1 hour) Studied the expected PS5 controller data stream for ESP32 integration.
 - Identified controller data mappings:
 - Left analog stick axis_x: -512 to 511
 - R2 throttle trigger: 0 to 1023
 - L2 brake trigger: 0 to 1023

Lili Nguyen (Total Time Spent: 3 hours)

- (1.5 hours) Organized team meetings to clarify project direction and assign upcoming work.
- (1 hour) Finalized the PDR, CDR, and final demo schedule.
- (0.5 hour) Created the previous week's WAR.
- (1 hour) Reviewed the DBW PCB and studied the first-generation circuit wiring diagram.

Plans for the Upcoming Week

Ousmane Segba Kone

- Set up the MQTT broker and verify publisher/subscriber communication.
- Implement Logitech G29 input capture on the laptop.
- Validate that control messages can be published to MQTT.
- Flash the ESP32 with subscriber code and begin bench testing.

Pundit Vorakitolan

- Test the driving simulator hardware.
- Finalize PDR diagrams, including the block diagram and system design.
- Identify latency concerns and possible optimization areas before the PDR.

Van Lang

- Connect the source code with the ESP32.
- Work with Ousmane on MQTT communication.
- Continue testing controller input mapping.

Lili Nguyen

- Confirm PDR block diagram deliverables and milestones.
- Prepare for the PDR review meeting.
- Look ahead to Critical Design phase planning.
- Keep WAR submission and team communication on schedule.
- Meet with the sponsor on Monday to provide updates and ask follow-up questions.

Problems/Questions Requiring Attention of Faculty/Sponsor

The team would like confirmation on the following items:

- Is CAN bus officially out of scope for this project?
- Is serial communication to the ESP32 the intended control method?
- What latency target should the team design for?
 - Minimum command rate
 - Acceptable end-to-end delay
- Are there any lab-specific requirements for:
 - The Logitech G29 input script
 - The control message format
 - E-stop handling
 - Communication-loss failsafe behavior

Week 2

2/23/26 to 3/1/26

Executive Summary

This week, the team made progress on the main control path for the teleoperation system. Python-to-MQTT publishing was implemented for Logitech G29 control messages, and ESP32 subscriber firmware was developed to receive JSON messages, parse the data, and map control values to PWM outputs.

The team also added important safety and performance features. A 250 ms communication-loss failsafe was added to help prevent runaway behavior, and a Python bridge was designed to support 100 Hz data streaming so the system can work toward staying under the 100 ms latency target. Since full hardware access was limited, a ghost simulator was created to generate test commands through MQTT.

Next week, the team will focus on testing the firmware on the physical ESP32 and Logitech G29, completing PWM mappings, measuring latency, and verifying E-stop behavior through Wi-Fi disconnect testing.

Progress Toward Objectives and Assignments Completed:

Ousmane Segba Kone (Total Time Spent: 3 hours)

- (1 hour) Developed the initial Logitech G29 input script, including Python environment setup and input reading.
- (0.5 hours) Added MQTT connection setup and message publishing logic.
- (1 hour) Developed ESP32 MQTT subscriber code in Arduino IDE.
- (0.5 hours) Reviewed code and prepared for device-based testing before vehicle testing.
- Structured control messages using JSON fields for:
 - Steering
 - Throttle
 - Brake
 - E-stop
- Mapped steering and throttle values to PWM signals for ESP32 output.

Pundit Vorakitolan (Total Time Spent: 5 hours)

- (3 hours) Prepared documentation to support the PDR.
- (2 hours) Researched communication strategy recommendations for future project improvements.
- Reviewed recommendations from the PDR.
- Began refining MQTT-related settings, including:
 - QoS
 - Timing intervals
 - Reliability considerations
- Prepared communication tradeoff recommendations:
 - 5G as a stronger short-term option
 - Starlink as a possible long-term option

Van Lang (Total Time Spent: 4 hours)

- (1 hour) Programmed the car_receiver.ino firmware, including callback functionality for JSON packets.
- (1 hour) Created a Python ghost-data generator to simulate steering wheel inputs and verify 100 Hz telemetry capability.
- (1 hour) Implemented 250 ms emergency-stop logic to prevent runaway behavior if the network disconnects.
- (1 hour) Checked code memory efficiency so additional protocols can be added later.
- Developed ESP32 failsafe logic in Arduino IDE.
- Created a Python bridge script for Logitech G29-to-MQTT communication.

Lili Nguyen (Total Time Spent: 3 hours)

- (1 hour) Organized a team check-in and assigned integration/testing tasks.
- (1 hour) Collected team progress updates and consolidated them into WAR format.
- (1 hour) Planned next-week milestones and prepared notes for faculty/sponsor updates.
- Coordinated team communication to keep the project pipeline organized.
- Communicated with other project managers about overall design.

Plans for the Upcoming Week

Ousmane Segba Kone

- Integrate Logitech G29 input capture with MQTT publishing.
- Validate that the live control message format stays consistent.
- Continue refining ESP32 PWM mappings.
- Confirm correct ESP32 response
- Support latency measurement and tuning across the full path:
 - Python publisher
 - MQTT broker
 - ESP32 subscriber
- Prepare for staged vehicle testing after bench validation.

Pundit Vorakitolan

- Test the driving simulator hardware.
- Continue refining MQTT recommendations for:
 - QoS
 - Timing intervals
 - Retransmission behavior
- Use integration test results to improve recommendations for latency and reliability.
- Prepare communication tradeoff recommendations for Dr. Xu.

Van Lang

- Validate that the firmware works on the ESP32.
- Connect the ESP32 system with the Logitech G29.
- Map the G29 control data to the PWM signals required by the car's steering system.
- Work with Ousmane to keep latency under 100 ms.
- Test the emergency stop by disconnecting Wi-Fi and verifying that vehicle control values default to zero.

Lili Nguyen

- Validate that the firmware works on the ESP32.
- Connect the ESP32 system with the Logitech G29.
- Map the G29 control data to the PWM signals required by the car's steering system.
- Work with Ousmane to keep latency under 100 ms.
- Test the emergency stop by disconnecting Wi-Fi and verifying that vehicle control values default to zero.

Problems/Questions Requiring Attention of Faculty/Sponsor

The team would like confirmation on the following items:

- What hardware interfacing constraints should the team follow?
 - Allowed PWM range
 - Steering neutral position
 - Steering and throttle limits to prevent mechanical damage
- Which ESP32 board revision should the team use?
- What cable is required for the ESP32?
 - Micro-USB
 - USB-C
- What network connection should the project use for testing and final design?
 - Wi-Fi
 - 5G
 - LTE
 - Starlink

Week 3

3/2/26 to 3/8/26

Executive Summary

This week, the team worked on reading Logitech G29 data, sending telemetry between laptops, and preparing for ESP32 integration. Van connected the Logitech G29, normalized the steering, throttle, and brake values, and added an emergency-stop feature. Ousmane created a subscriber program to receive and display live telemetry data. Pundit tested the driving simulator, collected calibration data, and worked on ESP32 setup. Lili coordinated team updates, meetings, and documentation.

Next week, the team will focus on fixing script crashes, lowering latency, continuing ESP32 setup, and moving toward car control.

Progress Toward Objectives and Assignments Completed:

Ousmane Segba Kone (Total Time Spent: 5 hours 15 minutes)

- (2 hours 15 minutes) Created a ZeroMQ subscriber and added topic/message parsing.
- (1 hour 30 minutes) Decoded telemetry data and added default values for missing fields.
- (1 hour 30 minutes) Displayed live steering, throttle, and brake values in the console.

Pundit Vorakitolan (Total Time Spent: 6 hours)

- (6 hours) Tested the driving simulator, collected calibration data, created test scripts, and worked on ESP32 configuration.

Van Lang (Total Time Spent: 5 hours)

- (2 hours) Worked with Ousmane to set up telemetry communication.
- (1.5 hours) Wrote code to normalize Logitech G29 steering, throttle, and brake values.
- (1.5 hours) Tested and troubleshooted the G29 interface.
- Added an emergency-stop feature so outputs go to 0 if the G29 disconnects.

Lili Nguyen (Total Time Spent: 5 hours)

- (30 minutes) Collected team progress updates.
- (1 hour) Organized weekly status notes, WAR content, and CDR PowerPoint material.
- (30 minutes) Coordinated next steps for integration and testing.
- (3 hours) Scheduled meetings and met in the lab to work on Logitech G29 communication with the laptop code.

Plans for the Upcoming Week

Ousmane Segba Kone

- Extend telemetry communication to the ESP32.
- Help connect Logitech G29 control to the car through the ESP32.
- Support hardware integration testing.

Pundit Vorakitolan

- Continue ESP32 setup.
- Support simulator and ESP32 integration.
- Help move calibration testing toward car control.

Van Lang

- Fix Python script crashes.
- Keep latency low between the Logitech G29 and laptop.
- Configure Arduino IDE for ESP32 firmware flashing.

Lili Nguyen

- Schedule three team meetings for ESP32 integration.
- Track progress on script stability, ESP32 setup, and telemetry.
- Coordinate communication during hardware and software integration.

Problems/Questions Requiring Attention of Faculty/Sponsor

- Need guidance on the best communication setup from laptop telemetry to ESP32 car control.
- Need confirmation on the correct ESP32 software and firmware setup.
- Need more information on latency and hardware limits for safe control.

Week 4 and 5

3/9/26 to 3/22/26

Combined report / spring break period

Executive Summary

Our team made good progress on communication between laptops and the ESP32. Van and Ousmane successfully established a local connection that allowed live steering wheel, throttle, and brake data from the Logitech G29 to be transmitted between laptops and forwarded to the ESP32. Pundit focused primarily on debugging the ESP32 wireless connection and developing test scripts. Some integration issues remain, particularly around ESP32 interfacing, wireless standardization, and implementation of control outputs for the car.

Progress Toward Objectives and Assignments Completed:

Ousmane Segba Kone (Total Time Spent: 3 hours)

- (1.5 hours) Developed Python publisher/subscriber and serial communication.
 - Sent G29 steering, throttle, and brake data from one laptop to another.
 - Used ZeroMQ to receive data on the second laptop.
 - Sent the data to the ESP32 through USB serial.
 - Formatted the data as steer, throttle, brake.
 - Added an "OK" confirmation from the ESP32.
- (1 hour) Wrote ESP32 Arduino code to:
 - Read data from the laptop.
 - Split the data into steering, throttle, and brake values.
 - Print received values for testing.
 - Send "OK" back to confirm data was received.
- (0.5 hours) Integrated and tested the laptop-to-laptop-to-ESP32 data path.
 - Researched the serial port "resource busy" issue.
 - Confirmed the correct process: upload ESP32 code first, then run Python.

Pundit Vorakitolan (Total Time Spent: 6 hours)

- (4 hours) Debugged the ESP32 wireless link and connection to the server.
- (2 hours) Wrote testing scripts.
 - Main blocker was the lack of standardization in the ESP32 wireless setup.

Van Lang (Total Time Spent: 2.5 hours)

- (1 hour) Established a local connection between Van's laptop and Ousmane's laptop.
- (1 hour) Debugged Python crashes caused by memory and driver limitations while capturing Logitech G29 data.
- (0.5 hours) Maintained low-latency data transfer without freezing the terminal.

Lili Nguyen (Total Time Spent: 3.5 hours)

- (0.5 hours) Collected weekly progress updates from team members.
- (0.5 hours) Prepared the Weekly Activity Report and maintained project documentation.
- (0.5 hours) Planned upcoming deliverables, including CDR preparation and slideshow material.
- (2 hours) Scheduled and hosted team meetings.

Plans for the Upcoming Week

Ousmane Segba Kone

- Find a way to interface ESP32 readings with the car.
- Send control readings from the ESP32 to the car.
- Convert steering, brake, and throttle data into car control outputs.

Pundit Vorakitolan

- Prepare for CDR.
- Integrate software and hardware work with the rest of the team.

Van Lang

- Transition the data stream so the ESP32 can connect to Wi-Fi and receive steering and pedal positions.
- Translate steering and pedal data into physical motor control outputs.
- Configure Arduino IDE to flash the ESP32 with C++ code.

Lili Nguyen

- Track CDR preparation for Wednesday.
- Organize presentation materials.
- Follow up on unresolved issues and create an action plan for the team.

Problems/Questions Requiring Attention of Faculty/Sponsor

- The team is making progress, but ESP32 integration and communication issues may affect CDR readiness.
- The team may need additional time, possibly one more week, to finalize preparation and make sure the design is well supported.
- Feedback on whether the current progress is sufficient for CDR expectations would be helpful.

Week 6

3/23/26 to 3/29/26

Executive Summary

This week was all about CDR prep and data testing. We spent most of our time finalizing our presentation slides, practicing our delivery, and cleaning up our system diagrams. On the technical side, we successfully set up timestamps to measure the latency between the driver workstation and the receiving terminal. We also worked through some connection bugs to ensure the link is stable. Moving into next week, we're shifting our focus from "planning" to "plugging things in" as we start integrating the ESP32 into the full driving pipeline.

Progress Toward Objectives and Assignments Completed:

Ousmane Segba Kone (Total Time Spent: 4 hours)

- (2 hours) Worked with Van on latency measurement between the sender laptop and receiver laptop.
- (2 hours) Troubleshoot connection issues, completed CDR slides, and prepared for the presentation.

Pundit Vorakitolan (Total Time Spent: 6 hours)

- (4 hours) Prepared for the CDR.
- Worked on diagrams and presentation practice.
- Recreated an ESP32-to-laptop test setup for additional validation.

Van Lang (Total Time Spent: 4 hours)

- (2 hours) Worked with Ousmane to build timestamps for measuring latency between the sender and receiver.
- (2 hours) Documented progress and prepared for the CDR presentation

Lili Nguyen (Total Time Spent: 5 hours)

- (4 hours) Prepared for the CDR by creating diagrams and practicing presentation slides.
- (1 hour) Organized team meetings and project documents.
- Created and organized the CDR slideshow.

Plans for the Upcoming Week

Ousmane Segba Kone

- Add the ESP32 to the driving pipeline.
- Test data flow through serial communication.

Pundit Vorakitolan

- Integrate software and hardware.
- Apply CDR recommendations.

Van Lang

- Build a handshake between the driver workstation and the ESP32.

Lili Nguyen

- Plan three meetings for debugging and hardware work.
- Help push integration of the vehicle hardware and code.
- Track team progress toward the final demonstration.

Problems/Questions Requiring Attention of Faculty/Sponsor

- The team would like feedback on the PDR/CDR materials.
- Are the system diagrams and presentation slides clearly communicating the latency benchmarks?
- Is the technical depth appropriate for the faculty review?
- What should be improved before the final demonstration presentation in May?

Week 7

3/30/26 to 4/5/26

Executive Summary

This week, the team focused on getting communication between the driving simulator and the ESP32 in the car. The team made progress setting up the control pipeline, including controller input, laptop communication, and wireless communication.

The main issue is that data is not updating in real time. This is slowing down testing and debugging. To help solve this, the team is also exploring a simpler ESP32-to-ESP32 wireless connection as an alternative.

Overall, basic functionality is in place, but the team needs to improve reliability and real-time performance moving forward.

Progress Toward Objectives and Assignments Completed:

Ousmane Segba Kone (Total Time Spent: 5 hours)

- (2 hours) Worked on Arduino code for receiving voltage values from the ESP32.
- (2 hours) Set up the remote driving pipeline from the laptop to the lab laptop and then to the ESP32.
- (1 hour) Tested and debugged the communication pipeline.

Pundit Vorakitolan (Total Time Spent: 4 hours)

- (4 hours) Debugged code for wireless communication between the driving simulator and the car.
- Worked on a parallel solution using direct ESP32-to-ESP32 communication.
- Focused on bypassing the car-side laptop and communicating directly with the ESP32.

Van Lang (Total Time Spent: 4.5 hours)

- (2 hours) Configured Arduino code to transfer data from Bluepad32 to Pygame.
- (1.5 hours) Troubleshoot data transmission from the driver workstation to the laptop connected to the ESP32 in the car.
- (1 hour) Debugged Pygame code and worked on integrating it with the ESP32 and hardware.
- Identified an issue where data only updates when the terminal is stopped and restarted.

Lili Nguyen (Total Time Spent: 6 hours)

- (1.5 hours) Coordinated team progress and collected debugging findings from different approaches.
- (3 hours) Tested team members' code and reviewed the hardware in the MiLa Lab.
- (1 hour) Reviewed the system architecture and identified integration points between the simulator, laptops, and ESP32.
- (0.5 hours) Prepared and organized the Weekly Activity Report.

Plans for the Upcoming Week

Ousmane Segba Kone

- Continue debugging why received data is stuck at 0.
- Continue working on transmitting data to the ESP32.

Pundit Vorakitolan

- Identify the correct IP address for the ESP32-to-ESP32 wireless link.
- Continue testing the direct ESP32 communication approach.

Van Lang

- Resolve the delayed data transmission issue.
- Continue refining the Pygame-to-ESP32 communication pipeline.
- Validate real-time control responsiveness.

Lili Nguyen

- Support integration between the planned approaches and the actual system architecture.
- Assist with debugging communication inconsistencies.
- Track team progress and keep the team aligned with project objectives.

Problems/Questions Requiring Attention of Faculty/Sponsor

- Data only updates when the terminal is restarted.
- The issue may be related to buffering, socket handling, or serial communication settings.
- We need guidance on best practices for real-time data streaming between systems.
- Having difficulty determining the correct IP addressing and communication setup for ESP32 wireless links in the lab environment.

Week 8

4/6/26 to 4/12/26

Executive Summary

This week, the team successfully established initial communication between the Logitech G29 steering wheel and the RC car. The car's tires are now able to respond to steering input from the G29. However, the steering movement is not fully aligned with the G29 yet, and latency is still the main issue.

Software work focused on improving real-time data transmission, while hardware work focused on checking wiring, validating steering response, and identifying possible causes of delay.

Progress Toward Objectives and Assignments Completed:

Ousmane Segba Kone (Total Time Spent: 3 hours)

- (2 hours) Debugged Arduino IDE code for sending G29 steering values to the car wheels in real time.
- (1 hour) Modified the sender laptop code to improve communication reliability.

Pundit Vorakitolan (Total Time Spent: 4 hours)

- (4 hours) Continue debugging code to wirelessly interface between the car and driving sim.
 - Edit the code so latency is lower
 - Assisted in validating the steering motor's physical movement when receiving signals from the G29 pipeline

Van Lang (Total Time Spent: 4.5 hours)

- (4.5 hours) We got the car's tires to move with respect to the Logitech G29. The movements on the car are not exactly the same as the movement on the G29. We got the tires moving, but the latency is still high

Lili Nguyen: (3.5 hours)

- (1.5 hours) Documented current hardware configuration and identified potential mechanical contributors to latency
- (1 hour) Coordinated team progress, consolidated updates, and maintained the project repository

- (1 hour) Managed communication with team members and ensured alignment on weekly objectives

Plans for the Upcoming Week

Ousmane Segba Kone

- Fine-tune steering left and right
- Create server to connect to allowing use with different wifis
- Fix latency of communication

Pundit Vorakitolan

- Collaborate with software members to compare expected vs. actual steering angles during testing
- Assist with integrating updated communication pipeline into the hardware system

Van Lang

- Continue refining the mapping between G29 steering input and car steering output
- Investigate software side contributors to latency

Lili Nguyen

- Coordinate next week's milestones, ensure task distribution, and prepare the next WAR
- Update hardware documentation as system changes are made

Problems/Questions Requiring Attention of Faculty/Sponsor

- We are planning to implement a server to allow operation across different Wi-Fi networks.
- Clarification is needed on any network restrictions or requirements from the sponsor's environment.

Week 9

4/13/26 to 4/19/26

Executive Summary

This week the team made progress in networking, steering control, and project coordination. Work included configuring an Oracle Cloud Infrastructure network server, refining wheel steering behavior and troubleshooting the drive-by-wire system, and discussing car control architecture and hosting considerations. Project meetings supported coordination of tasks, progress tracking, and alignment on next steps. Focus for next week is continued DBW troubleshooting, remote connectivity testing, subsystem integration, and continued iteration on vehicle control.

Progress Toward Objectives and Assignments Completed:

Ousmane Segba Kone (Total Time Spent: 2 hours)

- (2 hours) Work on fine-tuning the wheels steering i.e left to left and right to right with a wider angle
- Troubleshooting the DBW

Pundit Vorakitolan (Total Time Spent: 0.5 hours)

- (0.5 hour) talked to grad student about car control architecture
 - Unfortunately not much progress on my side - I barely missed the professor on Monday, but I did a little digging wrt advantages/disadvantages of host providers

Van Lang (Total Time Spent: 2 hours)

- (2 hours) This week I configured a network server using Oracle Cloud Infrastructure

Lili Nguyen (Total Time Spent: 2.5 hours)

- (2 hours) Attended project meetings with team members to discuss progress, coordinate tasks, and review next steps for DBW troubleshooting and system integration
- (0.5 hour) Reviewed project status, milestones

Plans for the Upcoming Week

Ousmane Segba Kone

- Troubleshooting DBW and try connecting remote with different wifis

Pundit Vorakitolan

- Continue iterating on car

Van Lang

- Setting up the Logitech G29 and ESP32 to communicate over this server, also ensuring that the two can communicate without being on the same Wi-Fi server

Lili Nguyen

- Continue coordinating project progress and tracking team deliverables
- Follow up on subsystem integration progress, including DBW troubleshooting and remote connectivity tasks

Problems/Questions Requiring Attention of Faculty/Sponsor

- Input on car control architecture decisions as subsystem integration progresses
- Are there concerns or best practices for using Oracle Cloud Infrastructure for remote communication in this application?

Week 10

4/20/26 to 4/26/26

Executive Summary

This week, the team tested the car and fixed an issue where the throttle and brake were reversed. A small steering issue was also noticed and will be adjusted. The team also finished organizing the code, documentation, and final report in preparation for the final demo.

Progress Toward Objectives and Assignments Completed:

Ousmane Segba Kone (Total Time Spent: 4 hours)

- (2 hour) Conducted a test drive to evaluate real time vehicle response
- (1 hour) Identified that throttle and brake values need to be inverted, and noted a potential steering bias requiring adjustment to the right
- (1 hour) Continued to test new code in the MiLa

Pundit Vorakitolan (Total Time Spent: 3 hours)

- (2 hours) messing around with the repo and understanding the current state of the code courtesy of the rest of the team.
- (1 hour) Making initial commits to the repo that Lili created; writing documentation and the user manual.

Van Lang (Total Time Spent: 3 hours)

- (1 hour) Worked on testing the car to validate steering, throttle, brakes, and latency
- (2 hours) Fixed the code since the logic was inverted for the throttle and brake pedals

Lili Nguyen (Total Time Spent: 3 hours)

- (1 hour) Coordinated team contributions and updated documents
- (1 hour) Organized and set up final repository on Git
- (1 hour) Finalized final report with all WARs this semester

Plans for the Upcoming Week

Ousmane Segba Kone

- Quantify control latency and response consistency after fixes
- Validate corrected throttle and brake mappings under repeated test conditions
- Prepare for final demonstration and report

Pundit Vorakitolan

- Make the car work

Van Lang

- Investigate and correct steering bias observed during test drive
- Support final system checks in preparation for demonstration

Lili Nguyen

- Refine the repository on GitHub
- Upload code and documents on flash drives by Thursday
- Prepare and practice for final demonstration on Friday afternoon
- Continue testing

Problems/Questions Requiring Attention of Faculty/Sponsor

- No major road blocks now.
- However, are there recommendations for improving calibration or control tuning before final evaluation